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AArh64 Memory & Cache System — Technical Reference Workbook

Chapter 8 — SMP Coherency, Inter-Core Visibility, and Boot-Sequence Requirements (Based on ARM64 Cortex-A72)

8.1 Scope

This chapter defines the architectural and microarchitectural rules governing **SMP coherency** and **inter-core visibility** on ARM64 Cortex-A72. It covers:

- Cluster-level coherency
- Inter-core data visibility
- Inter-core instruction visibility
- Boot-sequence requirements for secondary cores
- Required cache maintenance operations
- Required barrier semantics
- Cortex-A72 snoop and coherency behavior

This chapter is essential for OS kernels, hypervisors, bootloaders, and firmware responsible for multi-core bring-up.

8.2 SMP Coherency Model (ARMv8-A)

ARMv8-A defines a **multi-copy atomic**, **cache-coherent** model within a cluster.

8.2.1 Multi-Copy Atomicity

A write becomes visible to all cores **at the same instant** once it reaches PoC.

8.2.2 Cluster-Local Coherency

Within a Cortex-A72 cluster:

- L1D caches are private
- L2 is shared
- L2 maintains a duplicate directory
- Snoop logic ensures coherency
- All cores observe consistent memory at PoU and PoC

8.2.3 Inter-Cluster Coherency

Across clusters:

- Coherency is maintained only at PoC
- L2 caches are not shared
- Snoop traffic is routed through interconnect

8.3 Inter-Core Data Visibility

8.3.1 CPU0 Writes → CPU1 Reads

If CPU0 writes data that CPU1 must read:

- Dirty lines may reside in CPU0's L1D
- CPU1 cannot see these lines
- CPU1 observes memory only at PoC

Required sequence on CPU0:

```
DC   CVAC, xN
DSB  ISH
```

CPU1 may optionally execute:

```
DSB  ISH
```

8.3.2 CPU1 Writes → CPU0 Reads

Symmetric behavior. All cores observe data only after propagation to PoC.

8.4 Inter-Core Instruction Visibility

Instruction visibility requires propagation to PoU (L2).

8.4.1 CPU0 Modifies Instructions for CPU1

Required sequence on CPU0:

```
DC   CVAU, xN
IC   IVAU, xN
DSB  ISH
ISB
```

Required sequence on CPU1:

```
ISB
```

8.4.2 Why CPU1 Must Execute ISB

CPU1 may have:

- Stale instructions in L1I
- Stale instructions in front-end pipeline
- Prefetched instructions
- BTB predictions based on old code

ISB forces refetch from PoU.

8.5 Boot-Sequence Requirements for Secondary Cores

8.5.1 CPU0 Responsibilities

CPU0 must:

1. Prepare secondary core bootstrap code
2. Ensure instruction visibility
3. Ensure data visibility
4. Release secondary cores from holding pen

Required sequence:

```
DC   CVAU, bootstrap_region
IC   IVAU, bootstrap_region
DSB  ISH
ISB
```

8.5.2 CPU1 Responsibilities

CPU1 must:

1. Execute ISB
2. Fetch bootstrap code from PoU
3. Initialize its own MMU and caches
4. Join SMP coherency domain

Required sequence:

```
ISB
```

8.6 Snoop Behavior in SMP Context (Cortex-A72)

8.6.1 Duplicate Directory

L2 tracks which lines may be dirty in each core's L1D.

8.6.2 Snoop Requests

When CPU1 reads a line modified by CPU0:

1. L2 issues snoop to CPU0
2. CPU0 returns dirty data
3. L2 merges response
4. CPU1 receives updated data

8.6.3 Snoop Completion

Guaranteed only after:

```
DSB   ISH
```

8.7 SMP Barriers

8.7.1 DSB ISH

Required for:

- Completing DC operations
- Completing snoops
- Ensuring visibility across cores
- Ensuring writebacks reach PoC

8.7.2 DMB ISH

Required for:

- Ordering memory accesses
- Ensuring correct sequencing

- Not required for completion

8.7.3 ISB

Required for:

- Instruction visibility
- Pipeline flush
- BTB/return-stack synchronization

8.8 SMP Memory Ordering Rules

8.8.1 Loads and Stores

Loads and stores may be reordered unless:

- Region is Device-nGnRnE
- Barriers enforce ordering
- Shareability is Inner or Outer

8.8.2 Shareability

Normal memory must be marked **Inner-shareable** for SMP correctness.

8.8.3 Cacheability

Normal memory must be marked **cacheable** for performance and coherency.

8.9 SMP Coherency Failure Modes

Incorrect SMP configuration may cause:

- Stale instruction execution
- Stale data reads
- Missing DMA descriptors
- Incorrect interrupt controller state
- Secondary core boot failure
- Deadlocks due to missing DSB

8.10 Summary Table

Operation	CPU0 Action	CPU1 Action	Required Barriers
CPU0 → CPU1 data	DC CVAC	Optional DSB	DSB ISH
CPU0 → CPU1 instructions	DC CVAU + IC IVAU	ISB	DSB ISH + ISB
DMA → CPU	DC CIVAC	None	DSB ISH
CPU → DMA	DC CVAC	None	DSB ISH
Secondary core boot	DC CVAU + IC IVAU	ISB	DSB ISH + ISB

8.11 Exercises (Technical Reference)

Exercise 8.1 — Inter-Core Data Visibility

Explain why CPU1 cannot observe CPU0's dirty L1D lines without DC CVAC.

Exercise 8.2 — Instruction Visibility

Describe why CPU1 must execute ISB after CPU0 modifies bootstrap code.

Exercise 8.3 — Snoop Logic

Explain how duplicate directory snoops ensure inter-core coherency.

Exercise 8.4 — Boot Sequence

Describe the required sequence for CPU0 to safely release CPU1 from the holding pen.

Exercise 8.5 — Shareability

Explain why Normal memory must be marked Inner-shareable for SMP correctness.